Chapter 3

Reevaluating IFC and Comparing USD in the Context of the Built Environment

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Abstract

This study conducts a structured comparative analysis between Industry Foundation Classes (IFC) and Universal Scene Description (USD) to evaluate their respective capacities for supporting next-generation information workflows in the built environment. IFC has long been the cornerstone of open data interoperability in Building Information Modeling (BIM), yet its rigid schema, declarative semantics, and file-based federation model present limitations in addressing emerging needs for dynamic collaboration, granular data extensibility, and AI-integrated operations. In contrast, USD introduces a scenegraph-based architecture with non-destructive composition, layered editing, and schema modularity, originally developed for visual effects pipelines but increasingly recognized for its potential across domains such as robotics, geospatial modeling, and digital twin development. The analysis is framed across five analytical dimensions: data structure and hierarchy, identification and referencing, semantics and extensibility, composition logic and collaboration, and standardization and governance. Findings suggest that while IFC provides foundational stability and regulatory alignment, USD offers transformative capabilities for real-time coordination, semantic fluidity, and multi-author workflows. However, its effective adoption in AECO contexts requires the development of domain-specific schema extensions, quality validation protocols, and integration with existing BIM toolchains. Building on comparative insights between IFC and USD, this study recommends a hybrid strategy in which USD is positioned to complement IFC, enabling more robust interoperability across digital building ecosystems. Future research directions include the institutionalization of schema governance, AI-driven querying over USD-encoded data, and pilot implementations in complex infrastructure environments to assess performance, adaptability, and long-term sustainability.

Keywords: building information modeling (BIM), industry foundation classes (IFC), universal scene description (USD), comparative analysis, built environment, digital twin

1. Introduction

The Architecture, Engineering, Construction, and Operations (AECO) industry is undergoing a major transformation as digitalization reshapes how information is produced, shared, and governed across building lifecycles. The push toward data-centric

collaboration, semantic interoperability, and AI-integrated processes is accelerating the evolution of existing modeling standards. Building Information Modeling (BIM) remains a core methodology within this shift, supported primarily by the Industry Foundation Classes (IFC) schema. Despite its wide adoption and regulatory alignment, IFC faces persistent challenges in areas such as real-time model federation, extensibility, and graph-based data reasoning [1–3].

In parallel, Universal Scene Description (USD), originally developed for the film and animation industries, is being explored as an alternative framework for structured, layered, and scalable model composition [4, 5]. USD's scenegraph structure, non-destructive authoring model, and schema-level modularity introduce capabilities that align closely with emerging AECO needs. Studies in domains such as VFX, biomedical engineering, and robotics demonstrate its versatility in representing complex, semantically rich environments [6–9]. This cross-domain adaptability raises the question of whether USD can also serve as a foundational framework for AECO modeling ecosystems.

While previous works have explored USD's technical features and integration strategies in non-AECO contexts, comparative academic assessments of USD and IFC within AECO-specific workflows remain limited [10, 11]. This research aims to address that gap by investigating the architectural and functional suitability of USD as either a complementary or alternative modeling standard for AECO data environments. The study adopts a comparative analysis method supported by five analytical dimensions: data structure and hierarchy, identification and referencing, semantics and extensibility, composition logic and collaboration, and standardization and governance.

This study is organized into six main sections. Section 1 provides a foundational overview of IFC and USD. Section 2 presents the literature background and prior research. Section 3 outlines the research design and analytical dimensions. Section 4 provides detailed findings from the IFC–USD comparison. Section 5 discusses implications, challenges, and strategic opportunities for adopting USD in AECO workflows. Finally, Section 6 offers conclusions and structured recommendations for future research and practical implementation.

1.1 Industry foundation classes (IFC)

The Industry Foundation Classes (IFC) standard, governed by buildingSMART International, has long stood as the cornerstone of open data interoperability in BIM [3, 12]. Introduced in the mid-1990s, IFC was envisioned as a vendor-neutral specification for representing building and construction data across diverse software environments. As depicted in **Figure 1**, its development has followed a path of incremental enhancements, moving from rigid object classifications to more modular and extensible schemas, as reflected in the ongoing transition toward IFC5 [13–19].

At its core, IFC uses a federated data structure that combines models from different stakeholders using reference mechanisms instead of native compositional inheritance, as shown in **Figure 2**. Its structured, object-oriented inheritance model and large taxonomy of predefined entities and property sets are what make it strong. But these same strengths also show weaknesses: the rigid schema can make it hard to add new features dynamically, and changing existing entities often means changing the schema formally or adding custom property extensions.

Also, IFC's use of Global Unique Identifiers (GUIDs) to keep track of objects can make it harder to keep referential integrity, especially in workflows that have a lot of

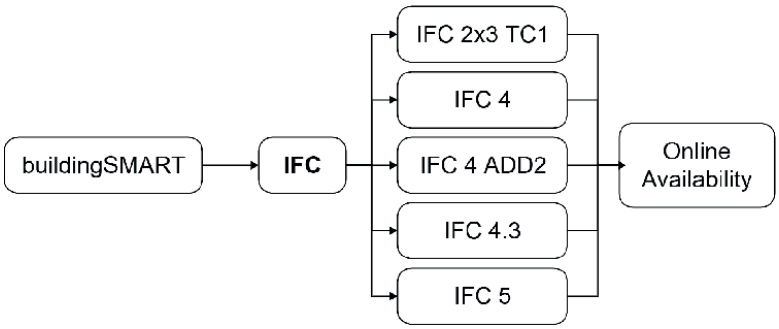


Figure 1.
IFC evolution and availability.

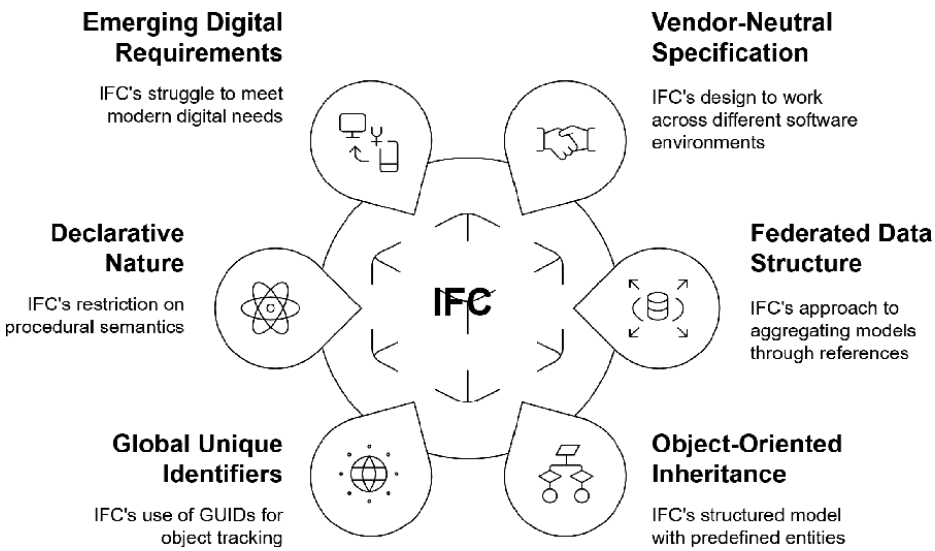


Figure 2.
Factors influencing IFC's evolution and limitations.

partial updates. The schema's declarative nature often makes procedural semantics less useful, which makes it less useful for modeling environments that use parameters or scripts.

IFC has been widely used, but its development has not kept up with new digital needs, especially in areas like real-time collaboration, data composability, and flexible schema overlays. The introduction of modular schema validation and frameworks like the BIM Collaboration Format (BCF) show that people are still trying to fill in these gaps, but the standard is still limited by its old architecture and governance model [20–28].

1.2 Universal scene description (USD)

Originally developed by Pixar to address the complex demands of film production pipelines, Universal Scene Description (USD) has since evolved into a robust framework for structuring, composing, and collaborating on complex 3D data, as shown in **Figure 3**. Its core objective was to allow multiple artists to work non-destructively

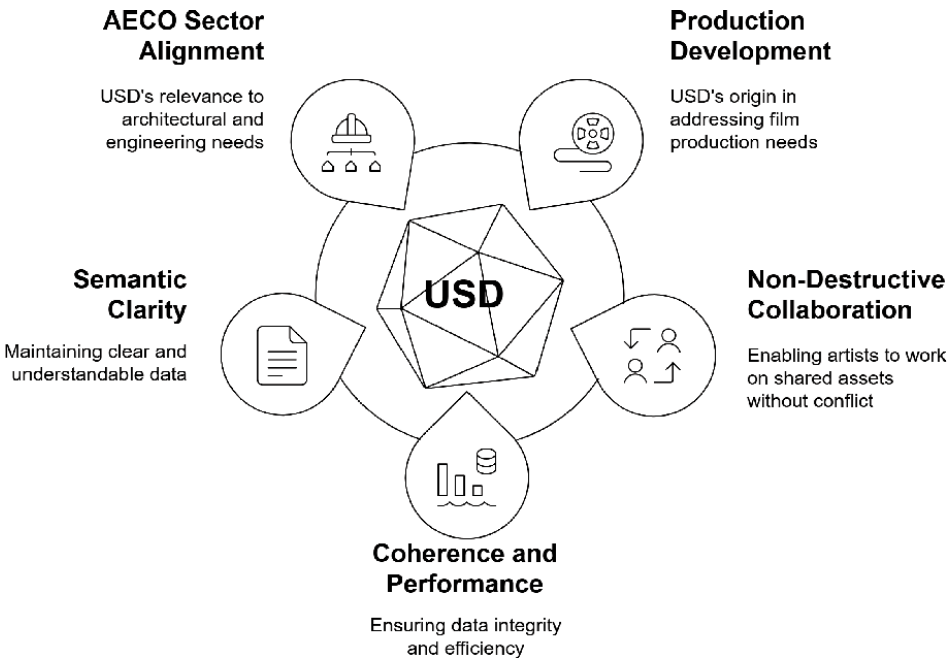


Figure 3.
Factors contributing to USD's robustness.

on shared digital assets across diverse tools and teams, while preserving coherence, performance, and semantic clarity. Although it originated within the entertainment industry, the architectural principles underlying USD closely align with the AECO sector's increasing requirements for model federation, traceability, and data extensibility.

At the core of USD lies the Stage-Prim-Layer hierarchy, as illustrated in **Figure 4**. The Stage functions as the runtime composition engine, aggregating multiple layers and references into a coherent model. Prims (primitive elements) are the fundamental

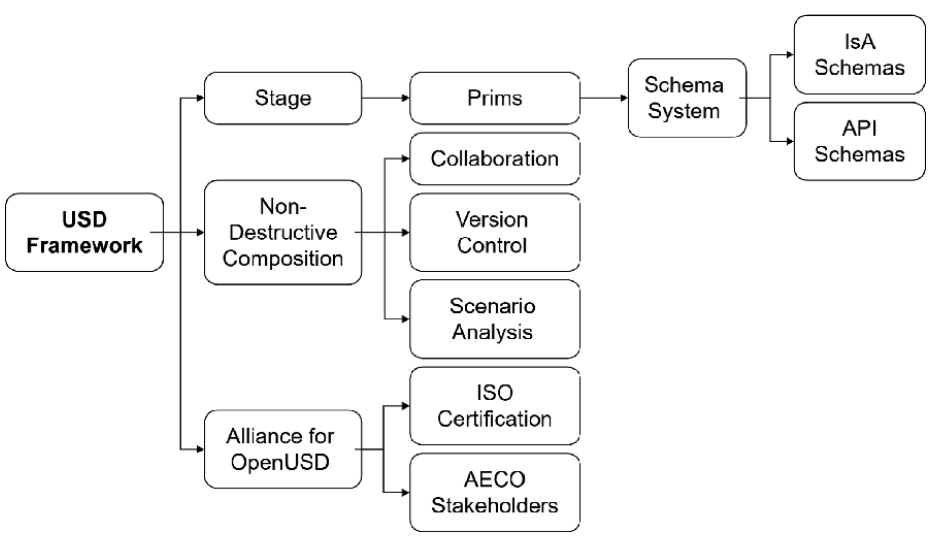


Figure 4.
USD framework and AECO integration.

units of content within a stage, analogous to objects or entities in BIM parlance. Unlike IFC's reliance on GUIDs, USD identifies these elements via namespace-based paths (e.g., Project/Site/Building/Room001/ExteriorWall003) favoring human readability and compositional clarity over strict identifier traceability.

As shown in **Figure 4**, USD's schema system offers two distinct approaches: IsA schemas define an object's type (e.g., UsdGeomMesh), whereas API schemas enrich objects with additional behaviors or metadata (e.g., UsdShadeMaterialBindingAPI). This distinction allows high degrees of modularity and domain specialization, which is critical for adapting a core framework to varied sectors like AECO, VFX, robotics, or gaming. Perhaps USD's most transformative feature is its non-destructive composition model. Using composition arcs such as reference, payload, variantset, and sublayer, users can layer modifications over base datasets without altering the underlying source. This model facilitates granular collaboration, version control, and scenario analysis, which directly align with BIM's emerging demands for partial model sharing, atomic transactions, and auditability.

Recognizing its cross-industry applicability, the Alliance for OpenUSD (AOUSD) was established in 2023 to oversee its governance, extension, and standardization [29]. The initiative is currently advancing toward ISO certification, and the inclusion of AECO stakeholders within AOUSD demonstrates a growing commitment to aligning USD's foundational capabilities with the specific needs of construction, operations, and facility management.

2. Literature review

The comparative analysis between IFC and USD within the AECO domain is an emergent area of inquiry. While IFC has a long-standing presence in BIM workflows, literature directly addressing USD's applicability in AECO remains limited but rapidly expanding, particularly through cross-domain insights. **Figure 5** presents a chronological distribution of key peer-reviewed publications. These works collectively explore core themes related to USD, including its architectural design, scene composition logic, and expanding relevance to AECO-oriented applications. As depicted in **Figure 6**, this review synthesizes key contributions grouped into three primary categories:

1. Core USD structure and workflows,

2. USD's potential in AECO-related digital ecosystems, and

3. Challenges and enablers for domain-specific adaptation.

2.1 Core USD structure and scene composition logic

Most of the early work on USD came from the digital content creation (DCC) and visual effects (VFX) fields. Bloom [10] gives a full picture of how USD is used in animation pipelines, with a focus on stage composition, schema layering, and non-destructive editing [10]. Bolstad [4] also looks at how USD can be used to manage large-scale cinematic visualizations, focusing on how hierarchical scene graphs work best when there are a lot of data [4]. Collins et al. [30] look into how well USD can scale up for use in large businesses, which shows how strong its architecture is [30].

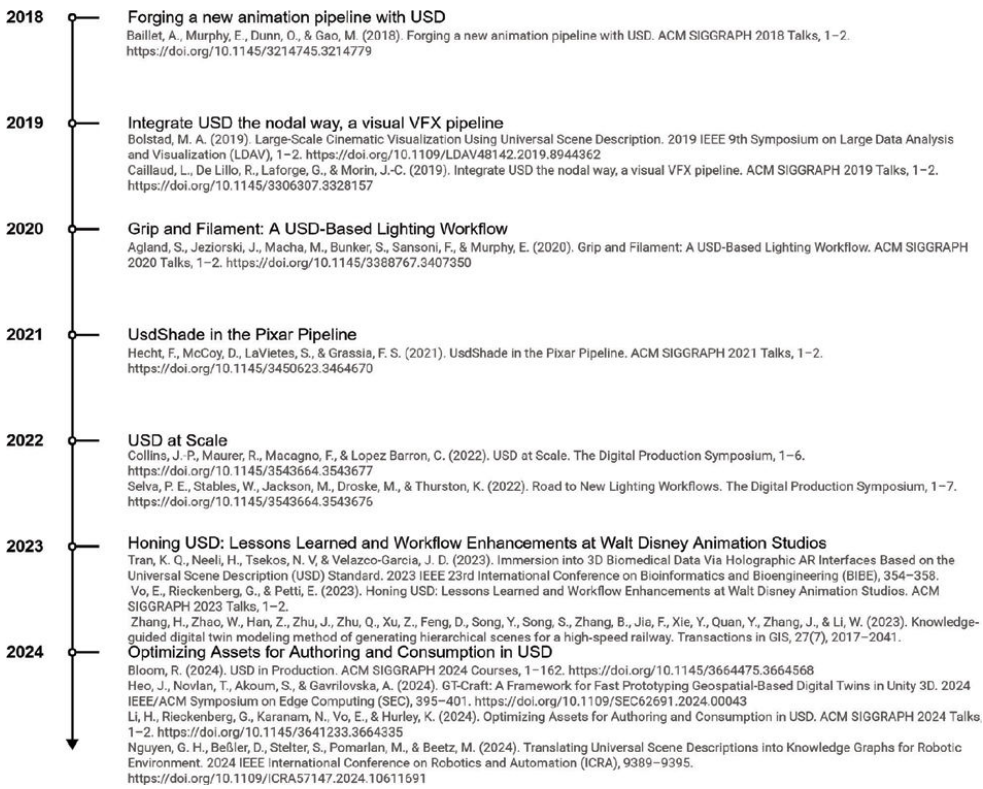


Figure 5.
Timeline of academic literature.

Characteristic	IFC	USD
Core structure	Long-standing presence in BIM workflows	Hierarchical scene graphs, schema layering
AECO-related digital ecosystems	Limited direct studies	Growing momentum through interdisciplinary research
Workflow integration	Transition to USD requires toolchain maturity	Lightweight asset referencing, scenegraph-based thinking

Figure 6.
IFC and USD comparison in AECO domain.

Caillaud et al. [31] look at graph-based nodal workflows, which gives us even more information about how flexible USD is for creating complex digital assets [31]. Vo et al. [5] and Li et al. [32] talk about how major studios improve and optimize their production processes. They show how USD makes it possible to create modular content with very little duplication [5, 32]. Hecht et al. [33] talk about the UsdShade schema and how it helps manage material definitions. They do this by showing how shader graph traversal works with USD's layered semantics [33]. When combined, these studies show how USD composition and schema management work on the inside. They also give us useful examples for how to structure AECO models.

2.2 USD in AECO-relevant digital ecosystems

There are not many studies that directly look at USD in the AECO field, but more and more research from different fields suggests that it could be useful. Zhang et al. [2] talk about a digital twin modeling method that is based on knowledge and uses hierarchical scene structures. These structures are very similar to USD's Stage–Prim organizational model [2]. Heo et al. [34] also introduce GT-Craft, a geospatial digital twin prototyping framework built in Unity 3D. It uses principles of modularity and composability that are very similar to USD's core architecture [34].

Nguyen et al. [9] make one of the most important contributions to AECO by turning USD structures into knowledge graphs that machines can understand [9]. This translation directly helps meet new needs for incorporating artificial intelligence (AI) and large language models (LLMs) into AECO model governance. Tran et al. [8] show how USD makes it possible to see biomedical models in 3D, which gives useful information for using spatialized representations in BIM workflows [8].

These studies show that USD originated in the entertainment industry, but its architectural features such as non-destructive layering, semantic schema extension, and namespace-based organization could prove highly beneficial for digital construction, operations, and infrastructure management.

2.3 Workflow integration, tooling, and adoption barriers

When moving from IFC-centric to USD-enhanced workflows, it is important to think about both the maturity of the toolchain and the best practices in the field. Baillet et al. [7], Agland et al. [35], and Selva et al. [11] all give real-life examples of how to integrate USD into production pipelines. They show the benefits of lightweight asset referencing and the problems that scenegraph-based authoring paradigms can cause [7, 11, 35]. These experiences give AECO professionals who are moving away from traditional federated modeling approaches useful information.

Kalyuzhnyy et al. [36] also show off an extensible DCC platform called "Wizart," which shows how USD-based tools can be made to work in specific fields [36]. Even though the focus is still on things other than construction, the technical foundation makes it possible for AECO-centric authoring environments.

3. Method

This study adopts a comparative analysis to examine the potential of Universal Scene Description (USD) as an alternative or complementary framework to Industry Foundation Classes (IFC) within the AECO sector as illustrated in **Figure 7**. The growing demands of digital construction, including composability, semantic governance, and AI-integrated workflows, highlight the need for a structured comparison that identifies how each modeling standard can address these challenges.

3.1 Research problem

IFC has long served as a foundational standard for data interoperability in BIM. However, its legacy structure has increasingly struggled to support newer demands such as real-time collaboration, graph-based composition, and flexible data overlays. On the other hand, USD, originally developed for the visual effects industry, offers

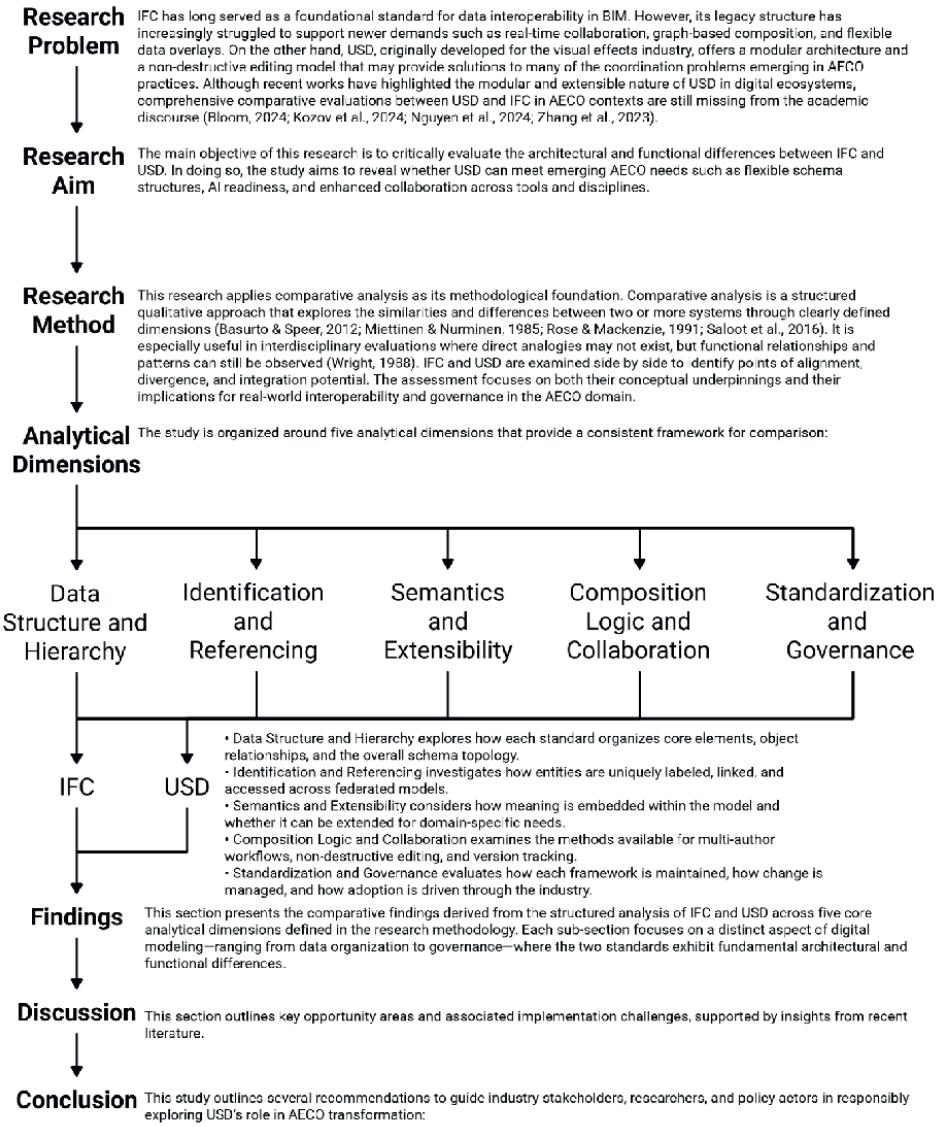


Figure 7.
Methodological framework for comparative analysis of IFC and USD.

a modular architecture and a non-destructive editing model that may provide solutions to many of the coordination problems emerging in AECO practices. Although recent works have highlighted the modular and extensible nature of USD in digital ecosystems, comprehensive comparative evaluations between USD and IFC in AECO contexts are still missing from the academic discourse [1, 2, 9, 10].

3.2 Research aim

The main objective of this research is to critically evaluate the architectural and functional differences between IFC and USD. In doing so, the study aims to reveal whether USD can meet emerging AECO needs such as flexible schema structures, AI readiness, and enhanced collaboration across tools and disciplines.

3.3 Research method

This research applies comparative analysis as its methodological foundation. Comparative analysis is a structured qualitative approach that explores the similarities and differences between two or more systems through clearly defined dimensions [37–40]. It is especially useful in interdisciplinary evaluations where direct analogies may not exist, but functional relationships and patterns can still be observed [41]. IFC and USD are examined side by side to identify points of alignment, divergence, and integration potential. The assessment focuses on both their conceptual underpinnings and their implications for real-world interoperability and governance in the AECO domain.

3.4 Research analytical dimensions

The study is organized around five analytical dimensions that provide a consistent framework for comparison:

- *Data structure and hierarchy* explores how each standard organizes core elements, object relationships, and the overall schema topology.
- *Identification and referencing* investigates how entities are uniquely labeled, linked, and accessed across federated models.
- *Semantics and extensibility* considers how meaning is embedded within the model and whether it can be extended for domain-specific needs.
- *Composition logic and collaboration* examines the methods available for multi-author workflows, non-destructive editing, and version tracking.
- *Standardization and governance* evaluates how each framework is maintained, how change is managed, and how adoption is driven through the industry.

These dimensions were selected based on their relevance to the practical concerns of AECO stakeholders and their significance in broader digital transformation goals.

4. Findings

This section presents the comparative findings derived from the structured analysis of IFC and USD across five core analytical dimensions defined in the research methodology. Each sub-section focuses on a distinct aspect of digital modeling, ranging from data organization to governance, in which the two standards exhibit fundamental architectural and functional differences. The analysis brings together both theoretical differences and real-world effects that are important for AECO workflows. It does this by using technical specifications, schema logic, and authoring models. **Table 1** gives a summary of these differences and explains what they mean for the AECO sector in different situations.

4.1 Data structure and hierarchy

Figure 8 shows how IFC organizes its data using a predefined schema made up of entities, relationships, and property sets that follow a strict inheritance-based

Analytical dimension	IFC (Industry Foundation Classes)	USD (Universal Scene Description)	Implication for AECO
1. Data structure and hierarchy	Rigid, inheritance-based schema with federated file structures. Object types are pre-defined and grouped via Property Sets. Models from different disciplines are externally referenced.	Flexible Stage-Prim-Layer architecture. Objects (*Prims*) are structured hierarchically and composed non-destructively across layers.	USD's flexible composition model could support more agile project phasing, partial updates, and better lifecycle integration in AECO workflows.
Example	*IfcWall* inherits from *IfcBuildingElement*; separate discipline models are merged manually.	*UsdGeomXform* exists in a stage, influenced by multiple authored or override layers.	Enables fine-grained object management and concurrent updates by different AECO stakeholders.
2. Identification and referencing	Uses globally unique identifiers (GUIDs) to track elements. Effective for long-term traceability but complicates incremental revisions and collaboration.	Identifies scene elements via hierarchical namespace paths (e.g., */Building/Floor2/RoomA/Wall03*), allowing more readable references.	USD's approach promotes better human readability and object traversal, especially in visual tools, though collision avoidance mechanisms would be needed.
Example	GUID: 3D02f9fB... assigned at creation; changes require regeneration and synchronization.	Path: /ProjectA/Site/BuildingA/Floor2/Window1 provides intuitive localization within the scene.	Hierarchical referencing could simplify facilities management systems or FM handover models based on spatial logic.
3. Semantics and extensibility	Semantics embedded in core schema; extension via Property Sets (PSets) is loosely governed and tool-dependent.	Separation of structural (*IsA*) and behavioral (*API*) schemas allows modular, scalable extensions. Localized behaviors can be added without modifying base definitions.	USD provides more sustainable extensibility, enabling AECO actors to introduce custom behaviors (e.g., fire rating, maintenance strategy) while preserving interoperability.
Example	*IfcDoor* can be extended with "Pset_DoorCommon," but validation consistency varies across tools.	A *UsdGeomMesh* can be augmented with *MaterialBindingAPI*, *VisibilityAPI*, or custom metadata schemas.	Enhances semantic clarity and adaptability for complex use cases like smart infrastructure or digital twin integration.
4. Composition and collaboration	Composition is file-based and external; disciplines author in silos and models are coordinated post-facto. Real-time collaboration is limited.	Layered, rule-driven composition supports non-destructive overrides, dynamic scenario testing, and simultaneous contributions across teams.	USD enables parallel, multi-disciplinary authoring and real-time what-if simulations, advancing AECO workflows toward digital twin maturity.
Example	Architects, MEP, and structural engineers submit separate models for coordination; integration is linear.	Designers can overlay energy analysis, code compliance, or simulation-specific layers without disrupting base geometry.	Supports more interactive design evaluation, sustainability assessment, and LOD-based phase planning.

Analytical dimension	IFC (Industry Foundation Classes)	USD (Universal Scene Description)	Implication for AECO
5. Standardization and governance	Long-established under buildingSMART with international recognition. Update cycles are formal but infrequent; adaptation to emerging tech like AI is slow.	Governed by AOUSD with a fast-moving open collaboration model. Actively evolving toward ISO certification and supports community-driven schema extensions.	AECO stakeholders can proactively shape USD standards to reflect sector-specific needs, ensuring earlier alignment with data governance and automation paradigms.
Example	IFC4.3 added infrastructure entities after years of consultation.	AOUSD announced domain working groups for manufacturing, VFX, and AECO, signaling intent for broader applicability.	Offers early engagement opportunities for AECO professionals to co-develop schema extensions aligned with digital construction goals.

Table 1.
Comparative analysis of IFC and USD across key analytical dimensions with AECO implications.

hierarchy. This structure helps with semantic clarity by allowing for type specialization and consistent classification across fields. But it also limits flexibility, especially when you need to change things at runtime, reuse parts of a model, or restructure things on the fly. The federated structure of IFC models encourages writing in a specific field and organizing files into modules by linking to them from other places. This method, on the other hand, can make it hard to integrate and coordinate data because it can cause problems with version updates or cross-disciplinary alignment.

USD, on the other hand, uses a Stage-Prim-Layer architecture, which is also shown in **Figure 8**. A Stage is like a scene graph that has Prims, which are the smallest parts of model content, organized in a hierarchy. Layers are used to write and change these Prims. Layers define both the original content and the contexts that override it. This architecture allows for high modularity and granular, non-destructive composition workflows. It also makes it possible to update things in real time and write things at the same time, which lets model changes spread dynamically without changing the base geometry or structure. In the AECO sector, where design changes, operational updates, and coordination between teams are becoming more common, these kinds of features are especially useful.

4.2 Identification and referencing

Figure 9 shows how IFC uses GUIDs to find and keep track of model elements. GUIDs are permanent references that work in distributed or federated BIM environments. This method helps keep data accurate and easy to check throughout the life of a project. However, it has some problems during iterative design revisions or collaborative authoring cycles, when updating models can cause GUIDs to change, which can make it harder to keep track of versions and break continuity. Also, the GUIDs themselves are not human-readable, which makes it harder to navigate scenes and refer to things in context.

USD, by contrast, utilizes a hierarchical namespace path system for object identification (e.g., Project/Site01/BuildingA/Floor2/StructuralWall03), also represented in **Figure 9**. This approach reflects the scene graph's structure and allows for direct traversal, semantic grouping, and visual interpretation of model hierarchies. Namespace paths are inherently more readable and align well with user interface

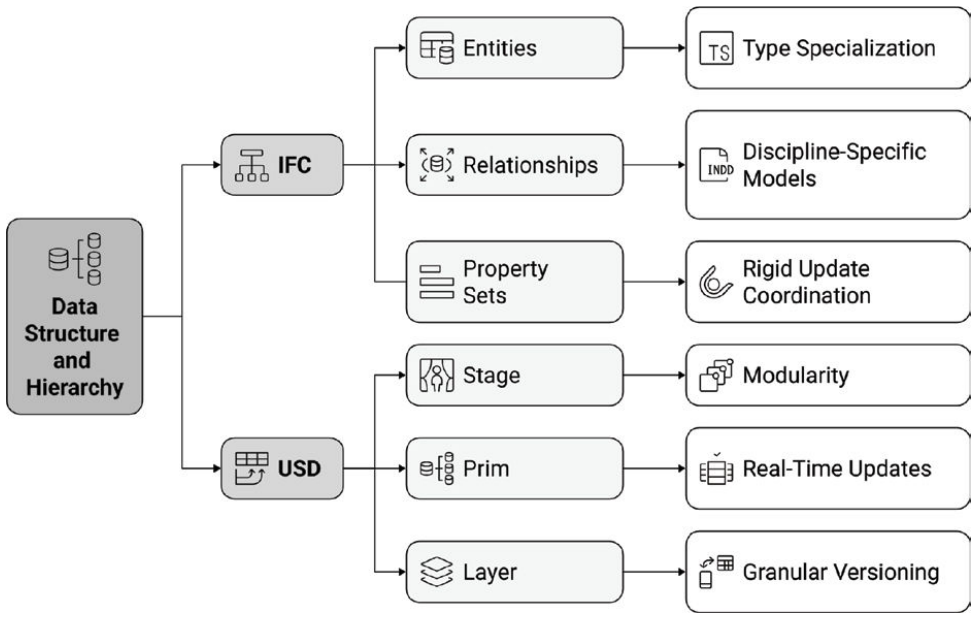


Figure 8.
Data structure and hierarchy comparison.

representations, particularly in environments that support tree-based navigation. Nonetheless, this flexibility introduces new challenges in large-scale, multi-author projects, where inconsistent naming or lack of namespace governance may result in collision risks or unintended overrides. These concerns highlight the importance of establishing clear authoring conventions and conflict-resolution mechanisms within collaborative AECO workflows.

4.3 Semantics and extensibility

In IFC, semantic meaning is embedded directly within its ontology-style schema, where entities encapsulate both structural definitions and domain-specific logic. This approach enriches information fidelity and consistency across disciplines. However, extensibility is primarily achieved through the use of custom Property Sets (PSets), which are loosely governed and inconsistently implemented across software platforms. As a result, interoperability and automated validation are often hindered, particularly when custom attributes lack schema-enforced definitions or standardized interpretation, as depicted in **Figure 10**.

USD adopts a more modular and granular approach by distinguishing between IsA schemas, which define the structural type of a primitive (e.g., UsdGeomMesh), and API schemas, which add optional behaviors or metadata (e.g., UsdShadeMaterialBindingAPI), as shown in **Figure 10**. This separation promotes composability and allows localized semantics to be layered onto base objects without requiring modifications to the core schema. The ability to incrementally extend functionality in this way supports diverse domain-specific requirements while maintaining consistency with the global data structure. For AECO applications, where models must balance compliance with standards and flexibility for project-specific constraints, USD's extensibility model offers a more sustainable and adaptable semantic framework.

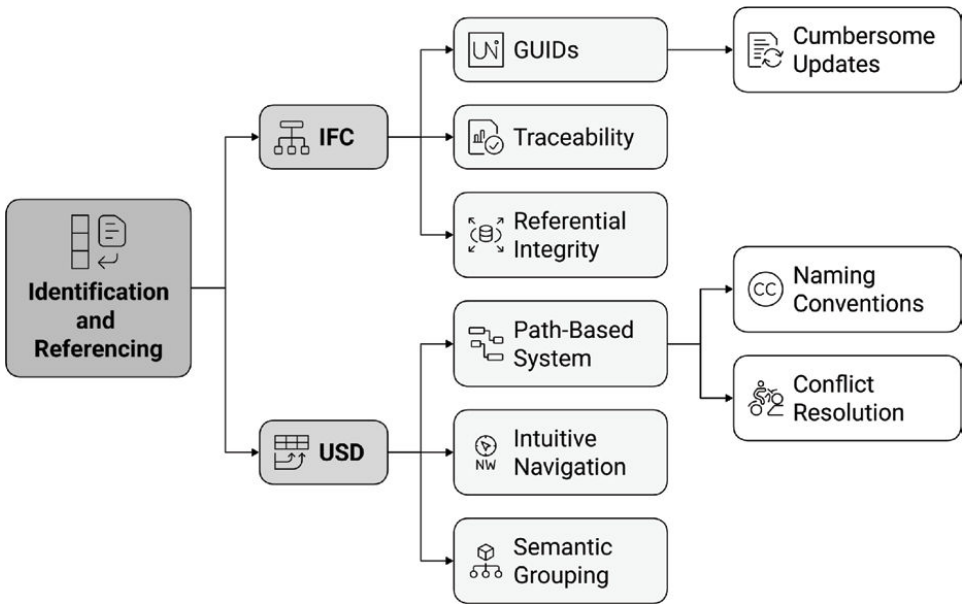


Figure 9.
Identification and referencing comparison.

4.4 Composition logic and collaboration

In IFC, model composition is achieved through a file-based federation approach in which discipline-specific models are authored independently and externally referenced before being aggregated during coordination. This sequential workflow supports formal design validation and clear responsibility separation. However, it introduces rigidity when managing iterative design changes, especially in projects that require frequent synchronization across teams or continuous integration of updates. The coordination model must be manually refreshed and reintegrated, which can become cumbersome in large-scale, fast-moving AECO projects. These limitations are conceptually illustrated in **Figure 11**.

USD employs a fundamentally different strategy based on layered, non-destructive composition. A scene can be constructed by referencing, overriding, or specializing content subsets using mechanisms such as reference, payload, and variantset. These composition arcs allow multiple contributors to work simultaneously on different aspects of the model without interfering with one another's content. Changes can be introduced through override layers that preserve the original structure, enabling seamless branching, rollback, and comparative evaluations. As depicted in **Figure 11**, this model supports scenario-based exploration, "what-if" simulations, and incremental authoring workflows. These capabilities align closely with digital twin development, integrated project delivery, and Level 3 BIM maturity targets.

4.5 Standardization and governance

IFC is maintained by buildingSMART International, a globally recognized standards organization that ensures consistency in documentation, version control, and cross-platform integration across the AECO industry. This institutional maturity provides long-term stability and broad adoption. However, its formal revision process

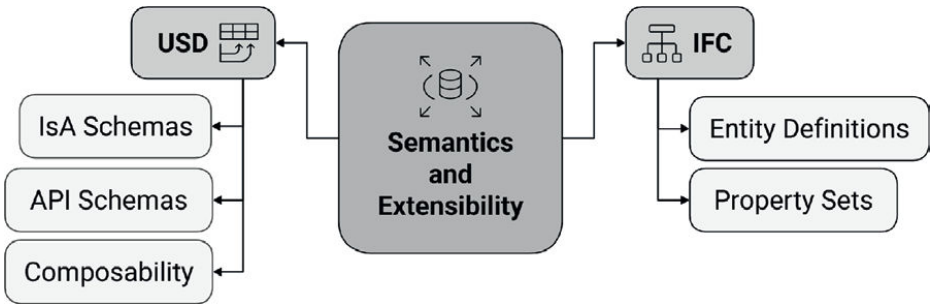


Figure 10.
Semantics and extensibility comparison.

tends to be slow, which can create challenges when adapting to emerging paradigms such as graph-oriented data modeling, semantic ontologies, or AI-driven information querying. These characteristics are illustrated in **Figure 12**.

USD, while relatively new, is governed by the Alliance for OpenUSD (AOUSD). This consortium includes a broad range of stakeholders in digital content creation and industrial technology who actively collaborate to evolve the standard. Its governance promotes transparent development and encourages domain-specific schema extensions. Ongoing efforts toward ISO certification and open tool alignment reflect this commitment. Although AECO-specific implementations and tooling are still developing, the growing participation of architects, engineers, and facility managers suggests that USD could be shaped to address the sector's operational, semantic, and lifecycle data needs. The governance approach and standardization pathway are visualized in **Figure 12**.

5. Discussion

The comparative findings presented in the previous section highlight that USD's foundational capabilities, including modular composition, semantic extensibility, and non-destructive collaboration, are closely aligned with AECO's shifting expectations

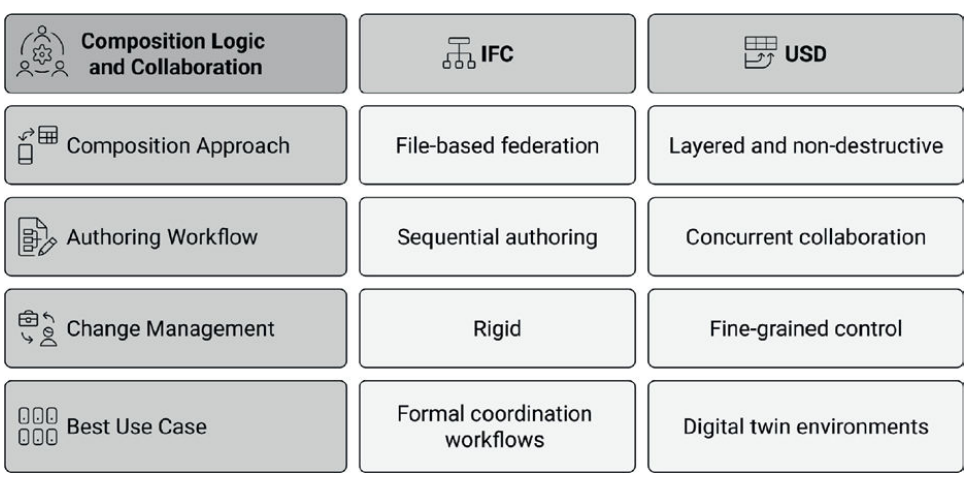


Figure 11.
Composition logic and collaboration comparison.

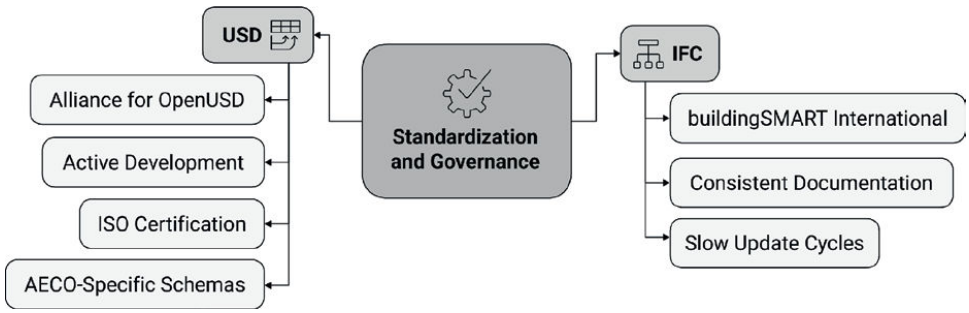


Figure 12.
Standardization and governance comparison.

in digital modeling and coordination. These attributes are particularly relevant in contexts requiring real-time authoring, federated model management, and adaptable data structures. However, realizing USD's full value in the AECO domain depends not only on technical potential but also on domain-specific adaptation, ecosystem maturity, and cultural readiness across stakeholders. As depicted in **Figure 13**, this section outlines key opportunity areas and associated implementation challenges, supported by insights from recent literature.

5.1 Domain adaptation and semantic localization

IFC embeds domain-specific semantics directly into its schema, enabling standardized classification across AECO workflows [3]. In contrast, USD offers a generic schema framework that remains agnostic of any single industry [10]. This structural neutrality allows AECO stakeholders to define localized schema packages, such as UsdAEC, which could encode architectural elements, construction sequences, and facility attributes in a way that aligns with project-specific semantics. Ongoing research emphasizes the value of schema-level localization to support modularity and reuse in AECO contexts [9].

5.2 LLM integration and graph-based reasoning

The scenegraph architecture of USD is a natural fit for a graph-based data structure made up of primitives and relationship arcs. This design fits well with recent improvements in large language models (LLMs) and graph neural networks that can understand how structured relationships work in design data [32]. Being able to show and move through spatial hierarchies, semantic groupings, and linked behaviors in a structured way opens up new possibilities for AI-assisted design intelligence, compliance checking, and automated coordination [9]. But these abilities will depend a lot on the creation of AECO-aligned vocabularies, graph grammars, and promptable ontologies that LLMs can understand in a meaningful way.

5.3 Validation, governance, and data assurance

IFC requires validation through XSD schemas and Model View Definitions (MVDs) to make sure that all regions are consistent and follow the rules. USD, on the other hand, uses a more decentralized approach [2]. Validation in USD is often defined through project-specific test scripts or custom grammars, offering

	Opportunities	Challenges
Domain Adaptation	Localized schema packages for AECO semantics	Requires domain-specific adaptation of generic schema
LLM Integration	AI-assisted design intelligence and automation	Development of AECO-aligned vocabularies needed
Validation & Governance	Flexibility through project-specific validation scripts	Risk of interpretive inconsistencies without shared protocols
Tooling & Ecosystem	Potential for domain-specific authoring tools	Lacks direct support for critical AECO functions
Cultural & Contractual	Distributed authoring and collaboration potential	Requires revision of contractual arrangements

Figure 13.
Opportunities and challenges for USD in AECO.

flexibility but requiring discipline to avoid interpretive inconsistencies [10]. To make sure that all teams and platforms follow the same rules and can work together, AECO stakeholders may need to set up shared validation protocols and reference implementations.

5.4 Tooling, training, and ecosystem maturity

USD's current tooling ecosystem, which grew out of animation and visual effects, does not directly support important AECO tasks like making 2D drawings, finding clashes, taking off quantities, and checking for code compliance [4]. To fill this gap, we need to make domain-specific authoring tools, visual editors, and integration platforms that work with BIM and CDE environments. Additionally, effective adoption will rely on training programs and professional development efforts that familiarize practitioners with the logic of scenegraphs, layering, and composition arcs [9, 10].

5.5 Cultural and contractual considerations

USD's support for layered authoring and decentralized collaboration creates a new way of thinking about model ownership and coordination. This method is different from the more linear and file-based methods that are common in current BIM standards. Because of this, current contracts may need to be changed to allow for the assignment of duties across overlapping layers, set access control rules, and keep track of who wrote what. Standards like ISO 19650 provide a basis for structured information delivery and responsibility matrices. These could be changed to show how USD can be used for distributed authoring [1].

6. Conclusion

This study conducted a structured comparative analysis to evaluate the suitability of USD as an alternative or complementary framework to IFC within AECO workflows. The research emerged from the limitations of existing standards in supporting

real-time coordination, semantic extensibility, and data-centric collaboration. To address this, the study used a qualitative comparison based on five analytical dimensions: data structure and hierarchy, identification and referencing, semantics and extensibility, composition and collaboration, and standardization and governance.

Findings demonstrated that USD's scenegraph-oriented architecture and layered composition model provide enhanced flexibility in authoring, federation, and semantic localization. Unlike IFC's rigid schema and file-based federation, USD enables non-destructive overrides, variant management, and domain-specific schema augmentation without altering the global structure. Moreover, USD's namespace-driven referencing system and dual-schema design (IsA and API schemas) align well with emerging applications involving AI-assisted modeling, data integration, and digital twin development.

Nevertheless, USD lacks a mature AECO-specific schema, standardized validation mechanisms, and integrated tooling support that reflect current industry workflows. While USD exhibits potential for interoperability and collaboration, its effective deployment in AECO requires intentional adaptation, stakeholder coordination, and cultural acceptance within established regulatory and contractual frameworks.

6.1 Recommendations for future research

As illustrated in **Figure 14**, and based on the synthesis of technical and organizational findings, this study outlines several recommendations to guide industry stakeholders, researchers, and policy actors in responsibly exploring USD's role in AECO transformation:

- *Pilot projects and prototyping*: Encourage cross-sector initiatives that prototype USD-based workflows in areas such as federated modeling, component authoring, and AI-enhanced compliance checking. Pilot studies will help validate USD's applicability, performance, and integration challenges under real-world constraints.

- *Development of AECO-specific schemas*: Promote collaborative efforts to establish domain-aligned USD schema packages, such as a modular UsdAEC extension.

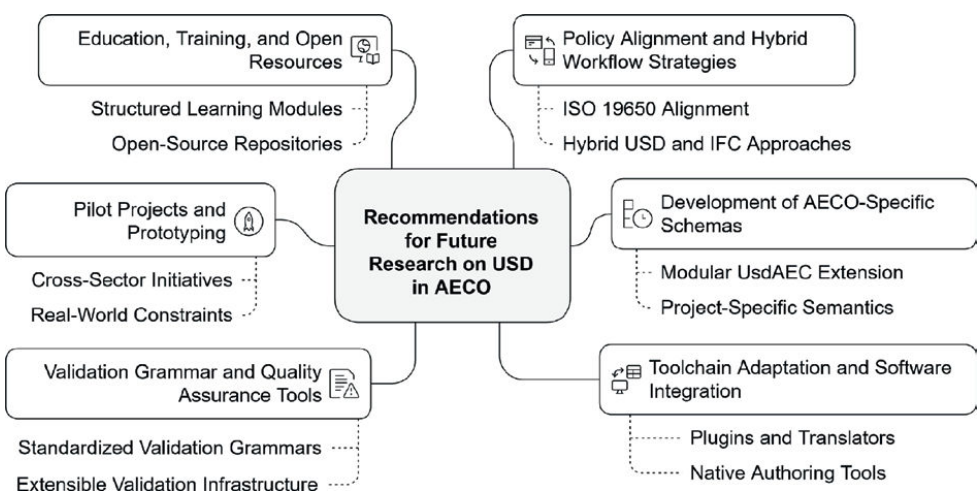


Figure 14.
Recommendations for future research on USD in AECO.

These efforts should capture architectural concepts, building systems, construction phases, and lifecycle attributes, fostering interoperability while preserving project-specific semantics.

- *Validation grammar and quality assurance tools*: Create standardized validation grammars and test suite templates to support regulatory and contract-driven conformance checking. These tools should reflect AECO requirements while leveraging USD's extensible validation infrastructure.

- *Toolchain adaptation and software integration*: Support the development of plugins, translators, and native authoring tools that bridge the gap between USD and common AECO platforms, including BIM authoring tools, CDEs, and Geographic Information System (GIS) systems.

- *Education, training, and open resources*: Design structured learning modules and open-source repositories to disseminate USD knowledge across disciplines. Educational initiatives should target architects, engineers, contractors, and data scientists to facilitate ecosystem-wide capacity building.

- *Policy alignment and hybrid workflow strategies*: Explore frameworks to align USD workflows with existing information management protocols such as ISO 19650 and the Common Data Environment (CDE). Hybrid approaches that blend USD and IFC may support incremental adoption and comparative evaluation of performance across domains.

In parallel with these recommendations, future research could focus on comparative simulations of IFC and USD in design coordination scenarios, the construction of USD-based knowledge graphs for semantic reasoning, and the assessment of long-term maintainability of graph-based models in facilities management. Longitudinal case studies involving public infrastructure or large-scale commercial projects would also help determine the operational feasibility and scalability of USD adoption within regulated AECO contexts.

By positioning USD not as a replacement but as a complementary modeling paradigm, this study advocates for a more flexible, intelligent, and collaborative foundation for next-generation AECO information management.

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Conflict of interest

The authors declare no conflict of interest.

Notes/thanks/other declarations

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Acronyms and abbreviations

AECO	architecture, engineering, construction, and operations
AI	artificial intelligence
AOUSD	alliance for OpenUSD
BCF	BIM collaboration format
BIM	building information modeling
CDE	common data environment
DCC	digital content creation
GIS	geographic information system
GUIDs	global unique identifiers
IFC	industry foundation classes
LLM	large language model
MVDs	model view definitions
PSets	property sets
USD	universal scene description
VFX	visual effects

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